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SPPU M.Sc. Data Science — Free Study Material

Python Programming for Data Science

DS-553-MJ

Interview Preparation — Topic-wise Q&A

Semester II · Core

Curated interview questions and answers for Python Programming for Data Science, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Python core

Q. List vs tuple vs set vs dictionary?

List: ordered, mutable. Tuple: ordered, immutable. Set: unordered, unique elements. Dictionary: key-value pairs, mutable.

Q. What are lambda, map, filter and reduce?

lambda creates anonymous functions; map applies a function to each item; filter keeps items passing a condition; reduce aggregates items to a single value.

Q. What is exception handling in Python?

Using try/except/else/finally to catch and handle runtime errors gracefully without crashing the program.

NumPy & Pandas

Q. Difference between a Python list and a NumPy array?

NumPy arrays are homogeneous, contiguous in memory and support vectorised operations, making them far faster for numeric computation.

Q. What is a Pandas DataFrame?

A 2-D labelled, tabular data structure with columns of potentially different types — the core object for data analysis in Pandas.

Q. How do you handle missing data in Pandas?

Detect with `isnull()`, remove with `dropna()`, or fill with `fillna()` using a constant, mean/median, or forward/backward fill.

Q. What does groupby do?

It splits data into groups, applies a function (e.g. aggregation) and combines the results — the split-apply-combine pattern.

